

Babar Hashmi

AI Engineer - Machine Learning Specialist

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Bangalore, India | Open to Relocation

Professional Summary

Results-driven AI Engineer with around 3 years of developing production-ready machine learning solutions, focusing on deep learning, agentic AI systems, and scalable model deployment. Proven track record delivering \$50K+ cost savings, 92% model accuracy, and 40% faster time-to-production. Expert in custom agent engineering, LLM fine-tuning, MLOps, and end-to-end ML pipelines. Founding AI R&D Hire (MTS) - Research Engineering track, building agentic systems 0 to 1.

Core Competencies

- > End-to-End MLOps: Taking models from research to production-grade deployment and monitoring.
- > Agentic AI System Design: Building tool-calling agents and multi-step reasoning loops.
- > Computer Vision Pipelines: Real-time detection systems for physical-world use cases.
- > LLM Fine-Tuning and Grounding: Improving contextual accuracy and eliminating hallucination.
- > SDK and API Engineering: Designing developer-facing tools adopted across business units.
- > Cross-Functional Delivery: Partnering with stakeholders from research through client rollout.

Professional Experience

AI Engineer - QuillAI Network (QuillAudits) | Jul 2024 - Aug 2026

- > Architected and deployed production deep learning models for smart contract vulnerability detection, improving audit accuracy by 25% and reducing manual review by 60%.
- > Engineered custom agentic AI systems using the Anthropic Claude SDK - designing tool-calling loops and agent harness architecture; shipped developer-facing SDK across 3 business units.
- > Partnered with C-level stakeholders on client-facing deployment and developer integration, delivering predictive analytics across business units.
- > Pioneered LLM fine-tuning achieving 25% improvement in contextual understanding, directly reducing customer escalations.

AI/ML Researcher - QuillAudits | Sep 2023 - Jun 2024

- > Spearheaded development of an AI-powered blockchain auditing tool as lead AI Engineer, increasing client engagement by 30%.
- > Implemented statistical modelling for blockchain security, processing 100K+ smart contracts and identifying 95% of vulnerabilities.
- > Conducted advanced data analysis for market due diligence, analysing \$50M+ investments using deep learning techniques.
- > Drove Agile delivery across cross-functional teams, improving delivery speed by 35%.

Key Projects and Achievements

MEP Industrial Elements Detection - Vision-Language Model Integration

- > Pioneered GroundingDINO integration developing custom computer vision pipeline achieving 85% detection accuracy across 1K+ site images.
- > Deployed MLOps infrastructure for real-time inference with AWS-based API, processing 500+ daily inspections.
- > Implemented model monitoring, reducing false positives by 40%.

Real-Time Pothole Detection System - Computer Vision and Production Infrastructure

- > Architected YOLOv8-based system with 92% accuracy, implementing real-time video analysis with REST APIs and AWS-based data management supporting municipal-scale deployment.
- > Built Streamlit interface with Apache Kafka streaming, achieving sub-5-second processing with 1000+ detections.

Financial Strategy Risk Scoring Engine - Deterministic, Fact-Grounded Risk Verdicts

- > Built a deterministic scorecard engine rendering risk verdicts for live investment strategies

from real, point-in-time market data across 7 independent risk axes - never a hand-written or model-generated verdict.

> Engineered a groundedness-gated LLM interface where the model only routes intent and phrases output in plain language; a verifier rejects the entire response if any claim can't be traced to source data.

> Validated the engine against real historical market collapses and shipped a one-command verification pipeline that regenerates and diffs every published number against committed evidence.

Technical Proficiencies

ML/AI Frameworks: TensorFlow, PyTorch, Transformers, Hugging Face, Anthropic SDK, custom agent frameworks. Computer Vision: YOLOv8, GroundingDINO, OpenCV, Vision-Language Models. Cloud and MLOps: AWS, OCI, Docker, CI/CD pipelines, model monitoring. Engineering and Dev Tools: TDD, agent harness design, SDK design, architecture review. Programming and Runtimes: Python, TypeScript, Bun, Solidity, SQL/NoSQL, Pandas, NumPy, Matplotlib, Plotly, Streamlit.

Certifications

> Agentic AI Certified Associate - Oracle - Jul 2026.

> Copado Certified AI - Copado - Jul 2026.

> OCI: GenAI Certified Professional - Oracle - Jul 2024.

> Prompt and AI Engineering Certification - MFT - Jun 2024.

Impact Snapshot

\$50K+ cost savings - 92% model accuracy - 25% LLM improvement - 60% faster manual review - Agentic AI systems delivered across 3 business units.

Domains of Expertise

> Blockchain and Web3 Security: Smart contract auditing, vulnerability detection, on-chain analytics.

> Construction Tech: Site inspection automation, MEP element detection.

> Public Infrastructure: Municipal-scale computer vision deployments.

> FinTech: Deterministic, fact-grounded risk-scoring pipelines.